# PHYSICS

1) In nuclear power plant, the energy released depends on the mass of uranium sample (m), length of oscillator ( ) and frequency (f) of oscillation as, $E = m^x \ell^y f^z$, then x + y + z = ?

(1) 1

(2) 5

(3) 2

(4) None of these

2)

From the point of view of significant figures, Which of the following is/are correct?
(i) 11.3 cm + 4 cm = 15.3 cm
(ii) 4.53 m – 1.2 m = 3.3 m
(iii) 5.45 kg – 3.2 kg = 2.25 kg
(iv) 84.8 cm + 48.6 cm = 133 cm

(1) (ii) only

(2) (i), (ii) and (iv) only

(3) (i), (iii) only

(4) (ii) and (iv) only

3) The linear mass density of a vibrating string is $1.3 \times 10^{-4}$ kg/m. A transverse wave is propagating on the string and is described by the equation y = 0.021 sin (x + 30t), where x and y are measured in meter and t in second. The tension in the string is approximately :–

(1) 0.12 N

(2) 0.48 N

(3) 1.20 N

(4) 4.80 N

4) A tuning fork produces four beats per second with two open organ pipe having length 30 cm and 31 cm. Find the frequency of tuning fork.

(1) 120 Hz

(2) 124 Hz

(3) 240 Hz

(4) 244 Hz

5) When a mass m is attached to a spring it extends by 0.2 m. The mass 'm' is given a slight additional extension and released, then find the time period $\left(\text{take } g = \pi^2\right)$.

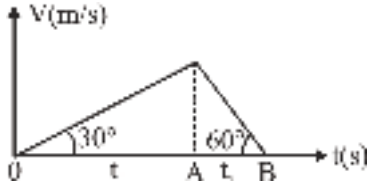
(1) $\frac{2}{5}$ sec

(2) $\frac{2}{\sqrt{5}}$ sec

(3) $\frac{5}{2}$ sec

(4) $\frac{5}{\sqrt{2}}$ sec

6) The velocity time graph of a body moving along a straight line is shown in figure. The ratio of the average velocities during the time $t_1$ and $t_2$ is :-




(1) 1 : 1

(2) 2 : 1

(3) 3 : 1

(4) 1 : 3

7) A bomber plane moves horizontally with a speed of 500 m/s and a bomb released from it, strikes the ground in 10 sec. Angle with horizontal at which it strikes the ground will be ($g = 10\ m/s^2$)

(1) $\tan^{-1}\left(\frac{1}{5}\right)$

(2) $\tan^{-1}\left(\frac{1}{2}\right)$

(3) $\tan^{-1}(1)$

(4) $\tan^{-1}(5)$

8) If $V_{rms}$ of $O_2$ molecule is equal to $V_{rms}$ of $H_2$ molecule at 20 K, the temperature of $O_2$ is :-

(1) 320°C

(2) 47 K

(3) 300 K

(4) 47°C

9) A pan filled with hot food cools from 94°C to 86°C in 2 minutes when the room temperature is 20°C. How long will it take to cool from 71°C to 69°C ?

(1) 10 s

(2) 12 s

(3) 24 s

(4) 42 s

10) A steel tape gives correct measurement at 20°C. A piece of wood is being measured with the

steel tape at 0°C. The reading is 25 cm on the tape, the real length of the given piece of wood must be:

(1) 25 cm

(2) <25 cm

(3) >25 cm

(4) can not say

11) Ball 1 collides with another identical ball at rest. For what value of coefficient of restitution e, the velocity of second ball becomes two times that of 1 after collision ?

(1) $\frac{1}{3}$

(2) $\frac{1}{2}$

(3) $\frac{1}{4}$

(4) $\frac{1}{6}$

12) A 500 kg boat is 9 m long and is floating without motion on still water. A man of mass 100 kg is at one end and if he runs to the other end of the boat and stops. The displacement of the boat is :

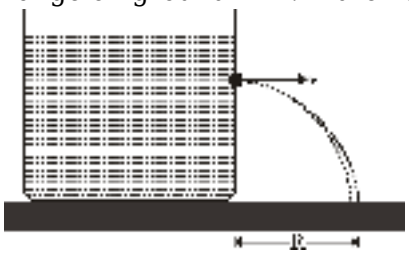
(1) 1.5 m in the direction of displacement of the man

(2) 0.75 m in the direction of displacement of the man

(3) 1.5 m in the direction opposite to the displacement of the man

(4) 0.75 m in the direction opposite to the displacement of the man

13) In the figure shown velocity of liquid which comes out is ν, time of liquid to fall to ground is t and range on ground in R. If the vessel is taken to a mountain, then match the following.



| | **Column I** | | **Column II** |
|---|---|---|---|
| (A) | ν | (P) | will increase |
| (B) | t | (Q) | will decreases |
| (C) | R | (R) | will remain same |

(1) A → P, B → Q, C → R

(2) A → Q, B → P, C → R

(3) A → R, B → P, C → Q

(4) None

14) An air bubble of radius 1.0 mm is observed at a depth of 20 cm below the free surface of water having surface tension 0.072 J/$m^2$ and density 1000 Kg/$m^3$. Then difference between pressure inside the bubble and atmospheric pressure is :-

(1) 144 N/$m^2$

(2) 72 N/$m^2$

(3) 2144 N/$m^2$

(4) 2288 N/$m^2$

15) If a surface is illuminated by a light of wavelength 2480Å and the maximum kinetic energy of emitted electrons is 2eV, then the work function ($\phi$) of the metal surface is:-

(1) 1eV

(2) 2eV

(3) 3eV

(4) 4eV

16) Match the corresponding entries of column-1 with column-2 (Where m is the magnification produced by the mirror) :-

| **Column-1** | | **Column-2** | |
|---|---|---|---|
| (A) | m = –2 | (a) | Convex mirror |
| (B) | $m = -\frac{1}{2}$ | (b) | Concave mirror |
| (C) | m = +2 | (c) | Real image |
| (D) | $m = +\frac{1}{2}$ | (d) | Virtual image |

(1) A → a and d, B → b and c, C → b and d, D → b and c

(2) A → a and c, B → a and d, C → a and b, D → c and d

(3) A → b and c, B → b and c, C → b and d, D → a and d.

(4) A → c and d, B → b and d, C → b and c, D → a and d

17) In a Young's double slit experiment the intensity at a point where the path difference is $\frac{\lambda}{6}$ ($\lambda$ being the wavelength of light used) is I. If $I^0$ denotes the maximum intensity, $\frac{I}{I_0}$ is equal to:-

(1) $\frac{3}{4}$

(2) $\frac{1}{\sqrt{2}}$

(3) $\frac{\sqrt{3}}{2}$

(4) $\frac{4}{3}$

18) The mass defect in a nucleus is 3.5 amu then the binding energy of the nucleus is :-

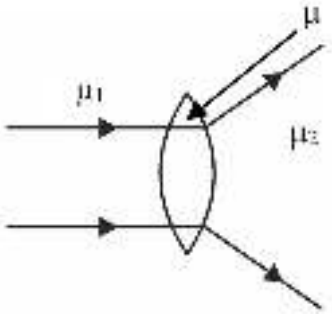
(1) 32.58 MeV
(2) 325.85 MeV
(3) 3260.25 MeV
(4) 3.258 MeV

19) A nucleus with mass number 240 breaks into two fragments of mass number 110 and 130 having binding energy per nucleon 7.8 MeV and 8.0 MeV respectively. If binding energy per nucleon of unfragmented nuclei is 7.5 MeV. The energy released in the process is :-

(1) 216 MeV
(2) 153 MeV
(3) 124 MeV
(4) 98 MeV

20) Light passes through a glass plate of thickness t and refractive index μ. For small angle of incidence (i) the lateral displacement is :-

(1) ti
(2) ti(μ – 1)
(3) $\frac{ti\mu}{(\mu-1)}$
(4) $ti\left(\frac{\mu-1}{\mu}\right)$




21) The correct conclusion that can be drawn from this figure is :

(1) $\mu_1 < \mu$ but $\mu < \mu_2$
(2) $\mu_1 > \mu$ but $\mu < \mu_2$
(3) $\mu_1 = \mu$ but $\mu < \mu_2$
(4) $\mu_1 = \mu$ but $\mu_2 < \mu$

22) A forward biased diode is.

(1) 
(2) 

(3) 

(4) 


23) An LED is constructed from a p-n junction diode using GaAsP. The energy gap is 1.9 eV. The wavelength of the light emitted will be equal to :-

(1) $10.4 \times 10^{-26}$ m

(2) 654 nm

(3) 654 Å

(4) $654 \times 10^{-11}$ m

24)

Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
**Assertion (A) :** In a conical pendulum, angular momentum about its vertical axis remains constant
**Reason (R) :** Net torque about vertical axis of conical pendulum is not zero.
In the light of the above statements, choose the most appropriate answer from the options given below:

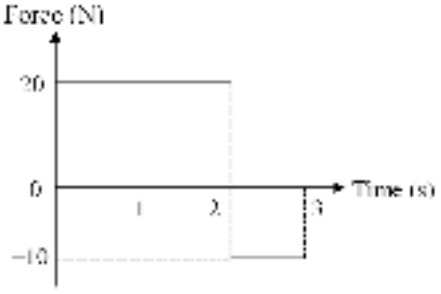
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true but **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

25) Only one force is acting on a 5 kg object which is at rest initially as indicated in the figure. Find the total work done by force :-




(1) 90 J

(2) 120 J

(3) 180 J

(4) 60 J

26) The maximum elongation of a copper wire of 10 m length if the elastic limit of copper and its young's modulus respectively are $1.5 \times 10^8$ N/m$^2$ and $12 \times 10^{10}$ N/m$^2$ is-

(1) 1.25 cm

(2) 12.5 cm

(3) 125 cm

(4) 12.5 m

27) If two bodies A and B of equal masses M are situated in air at a distance d and gravitational force between them is F. Now 50% mass is transferred from body A to B and distance between them is reduced to $\frac{d}{2}$. If space around them is now filled with a liquid of specific density 3. The gravitational force now will be :-

(1) F

(2) 3F

(3) $\frac{3F}{2}$

(4) $\frac{3F}{4}$

28) A body crosses the topmost point of a vertical circle with critical speed. Its centripetal acceleration, when the string is horizontal will be :-

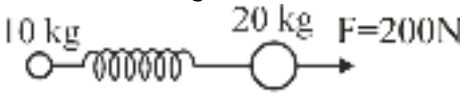
(1) 6g

(2) 3g

(3) 2g

(4) g

29) Two masses of 10 kg and 20 kg respectively are connected by a massless spring as shown in figure. A force of 200 N acts on the 20 kg mass. At the instant when the 10 kg mass has an acceleration of 12 $ms^{-2}$, the acceleration of the 20 kg mass is :



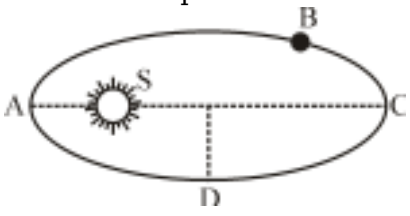

(1) 2 $ms^{-2}$

(2) 4 $ms^{-2}$

(3) 10 $ms^{-2}$

(4) 20 $ms^{-2}$

30) A planet revolves in elliptical orbit around the sun. The linear speed of the planet will be maximum at :




(1) A

(2) B

(3) C

(4) D

31) A particle of mass m = 5 units is moving with a uniform speed $\nu = 3\sqrt{2}$ unit in the XOY plane along the line Y = X + 4. The magnitude of the angular momentum of the particle about the origin is :

(1) Zero

(2) 60 unit

(3) 7.5 unit

(4) $40\sqrt{2}$ unit

32) A body of mass M is dropped from a height h on a sand floor. If the body penetrates x cm into the sand, the average resistance offered by the sand to the body is :

(1) $Mg\left(\frac{h}{x}\right)$

(2) $Mg\left(1+\frac{h}{x}\right)$

(3) Mgh + Mgx

(4) $Mg\left(1-\frac{h}{x}\right)$

33) A cylindrical region contains a uniform magnetic field that is along the cylinder axis and is changing with time. If r is distance from the cylinder axis the magnitude of the electric field within the region is :

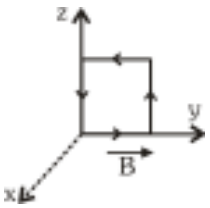
(1) uniform

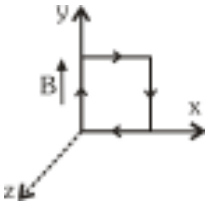
(2) proportional to 1/r

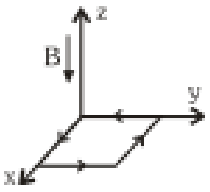
(3) proportional to $r^2$

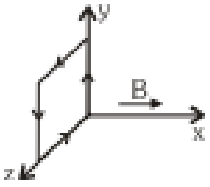
(4) proportional to r

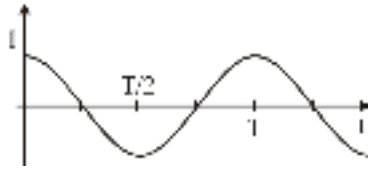
34) Which of the following loop is in stable equilibrium

(1) 

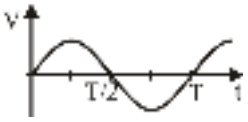

(2) 

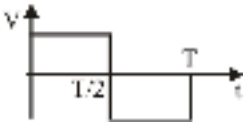

(3) 

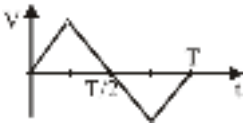

(4) 

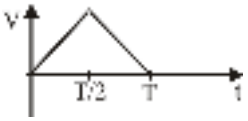
35) The current (I) in the inductance is varying sinusoidally with time according to the plot shown in figure. 

Which one of the following is the correct variation of voltage with time in the coil ?

(1) 


(2) 


(3) 


(4) 


36) The correct order of ionizing capacity of α, β and γ-rays is :-

(1) α > γ > β
(2) α > β > γ
(3) α < β < γ
(4) α < γ < β

37) A transformer is used to light 140 watt 24 volt lamp from 240 volt AC mains; the current in the main cable is 0.7 amp. The efficiency of the transformer is :

(1) 63.8%
(2) 94%
(3) 83.3%
(4) 48%

38) Power delivered by the ac source to the circuit becomes maximum, when :-

(1) $\omega L = \omega C$
(2) $\omega L = \frac{1}{\omega C}$
(3) $\omega L = -\left(\frac{1}{\omega C}\right)^2$
(4) $\omega L = \sqrt{\omega C}$

39) The magnetic field part of an EMW in a medium is $B_x = 0$, $B_y = 0$,

$B_z = 5\sin\left(\frac{\pi}{100}y + \pi \times 10^{+6}t\right) T$ then

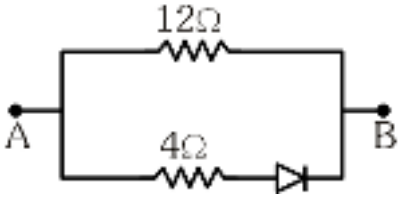
(1) It is moving in +y direction

(2) It is moving in –y direction

(3) It is of wavelength 100 m

(4) It is a frequency $5 \times 10^6$ Hz

40) Find the equivalent resistance of the network shown in the figure between the points A and B, if




(i) $V_A > V_B$ (ii) $V_A < V_B$

(1) 4Ω, 12Ω

(2) 3Ω, 4Ω

(3) 3Ω, 12Ω

(4) 12Ω, 3Ω

41) **Assertion :** No work is required to be done to move a test charge between any two points on an equipotential surface.
**Reason :** Electric lines of force at the equipotential surfaces are mutually perpendicular to each other.

(1) Assertion and Reason both are true, and Reason is the correct explanation of Assertion.

(2) Assertion and Reason both are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true and Reason is false.

(4) Assertion is false and Reason is true.

42) A resistance wire connected in the left gap of a metre bridge balances a 10Ω resistance in the right gap at a point which divides the bridge wire in the ratio of 4 : 5. If the length of the resistance wire is 2.0 m, then the length of 1Ω of the resistance wire is :-

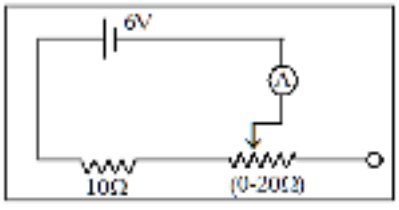
(1) $1.0 \times 10^{-1}$ m

(2) $1.5 \times 10^{-2}$ m

(3) $2.5 \times 10^{-1}$ m

(4) $1.0 \times 10^{-2}$ m

43) The resistance of the rheostat shown in fig. is 20Ω. The maximum & minimum current read by Ammeter is (Respectively) :-



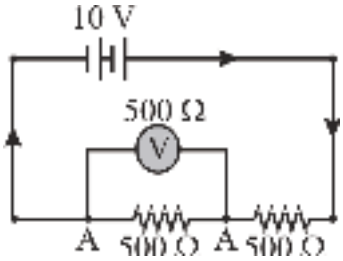

(1) 0.6 A, 0.3 A

(2) 0.3 A, 0.6 A

(3) 0.1 A, 0.3 A

(4) 0.6 A, 0.2 A

44) What is the reading of voltmeter in the following figure ?




(1) 3V

(2) 2V

(3) 5V

(4) $\frac{10}{3}$ V

45) How many capacitors each of 8μF and 250V are required to form a composite capacitor of 16μF and 1000V :-

(1) 8

(2) 32

(3) 16

(4) 64

## CHEMISTRY

1) Which of following have maximum ionisation energy :-

(1) $1s^2 2s^2 2p^3$

(2) $1s^2 2s^2 2p^6 3s^2 3p^3$

(3) $1s^2 2s^2 2p^6 3s^2 3p^5$

(4) $1s^2 2s^2 2p^5$

2) **Assertion :** Generally ionic compounds have high boiling point and melting point.
**Reason :** Due to strong electrostatic force of attraction among oppositely charged ions.

(1) Assertion and Reason both are correct and reason is correct explanation of Assertion.

(2) Assertion and Reason both are correct but reason is not correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

3) Which complex ion of cobalt will absorb light of highest wavelength ?

(1) $[Co(NH_3)_6]^{+3}$

(2) $[Co(CN)_6]^{-3}$

(3) $[CoF_6]^{-3}$

(4) $[Co(H_2O)_6]^{+3}$

4) Which of the following does not have S-S bond :

(1) $S_2O_4^{2-}$

(2) $S_2O_3^{2-}$

(3) $S_2O_5^{2-}$

(4) $S_2O_7^{2-}$

5) The group reagent for the IV group radicals is :

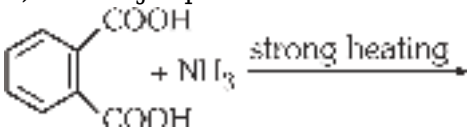
(1) $H_2S$ (acidic)

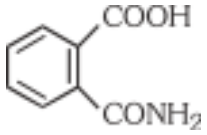
(2) NaOH

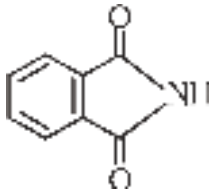
(3) $H_2S$ (basic)

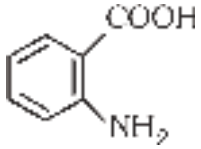
(4) $(NH_4)_2CO_3$

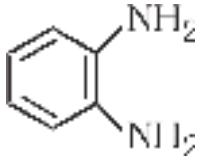
6) The major product of the following reaction is :




(1) 


(2) 


(3) 


(4) 


7) In the Cannizzaro reaction given below :

$$2Ph\text{–}CHO \xrightarrow[\Delta]{NaOH} Ph\text{–}CH_2OH + PhCOONa$$

the slowest step is :-

(1) The attack of $OH^-$ at the carbonyl group

(2) The transfer of hydride ion to the carbonyl group

(3) The abstraction of proton from the carboxylic acid

(4) The deprotonation of Ph–$CH_2OH$

8) $AB_2$ dissociates as
$AB_2(g) \rightleftharpoons AB(g) + B(g)$
when the initial pressure of $AB_2$ is 400 mm of Hg and the total equilibrium pressure is 600 mm of Hg. Then the value of $K_p$ is : (in mm Hg)

(1) 50

(2) 100

(3) 40

(4) 200

9) Consider the following statements :
(a) The half cell reaction taking place at the anode of Daniel cell is $Zn \rightarrow Zn^{+2} + 2e^-$
(b) Salt bridge provides an electric contact between the two solutions without allowing them to mix with each other
(c) By convention, a negative E° value means that the redox couple is weaker reducing agent than the $H^+/H_2$ couple.
Choose the incorrect statement(s) :

(1) a & b only

(2) b and c only

(3) a only

(4) c only

10) Which of the following statements concerning the quantum numbers are incorrect ?
(a) Orbital angular momentum quantum number determines the three dimensional shape of the orbital
(b) The principal quantum number determines the orientation and energy of the orbital.
(c) Magnetic quantum number determines the size of the orbital.
(d) Spin quantum number of an electron determines the orientation of the spin of electron relative to the chosen axis.

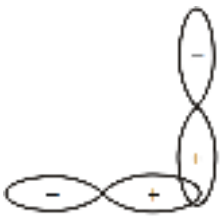
(1) a, b, c

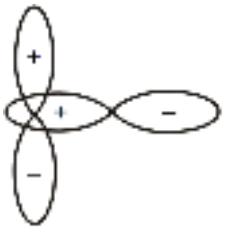
(2) b, d

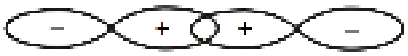
(3) b, c

(4) only b

11) Which p orbital overlapping would give the strongest bond ?

(1) 

(2) 

(3) 

(4) 

12) $N_2$ and $O_2$ are converted into monoanions, $N_2^-$ and $O_2^-$ respectively. Which of the following statements is wrong ?

(1) In $N_2^-$ , N-N bond weakens

(2) In $O_2^-$, O-O bond order increases

(3) In $O_2^-$, O-O bond order decreases

(4) $N_2^-$ becomes paramagnetic

13) **Assertion (A) :-** Mo (VI) and W (VI) are found to be more stable than Cr (VI).
**Reason (R) :-** Cr(VI) in the form of dichromate in acidic medium act as a strong reducing agent.

(1) A and R are both true but R is not correct explanation of A.

(2) A is true but R is false.

(3) A is false but R is true.

(4) A and R are both true and R is the correct explanation of A.

14) Number of structural isomers only having three membered ring from the molecular formula $C_6H_{12}$ ?

(1) 6

(2) 5

(3) 4

(4) 7

15) A $\xrightarrow[+CaO,\ \Delta]{NaOH}$ 2,2-Dimethyl butane; **A** is :-

(1) 3,3-Dimethyl pentanoic acid

(2) 4,4-Dimethyl pentanoic acid

(3) 2,3,3-Trimethyl butanoic acid

(4) All of above

16) Which of the following will undergo Hell-Volhard Zelinsky reaction?

(1) HCOOH

(2) $(CH_3)_3CCOOH$

(3) $C_6H_5COOH$

(4) $CH_3COOH$

17) If for a certain reaction $\Delta_rH$ is 50 kJ $mol^{-1}$ at 450 K, the value of $\Delta_rS$ for which the reaction will be spontaneous at the same temperature is :

(1) –100 J $K^{-1}$ $mol^{-1}$

(2) 120 J $K^{-1}$ $mol^{-1}$

(3) –56 J $K^{-1}$ $mol^{-1}$

(4) 79 J $K^{-1}$ $mol^{-1}$

18) The number of radial nodes and total number of nodes in 3d orbital respectively are :

(1) 1 and 2

(2) 0 and 1

(3) 0 and 2

(4) 1 and 1

19) 1.00 g of a non electrolyte solute dissolved in 50 g of benzene lowered the freezing point of benzene by 0.40 K. The freezing point depression constant of benzene is 5.12 K kg $mol^{-1}$. Find molar mass of solute.

(1) 128 g $mol^{-1}$

(2) 256 g $mol^{-1}$

(3) 64 g $mol^{-1}$

(4) 200 g $mol^{-1}$

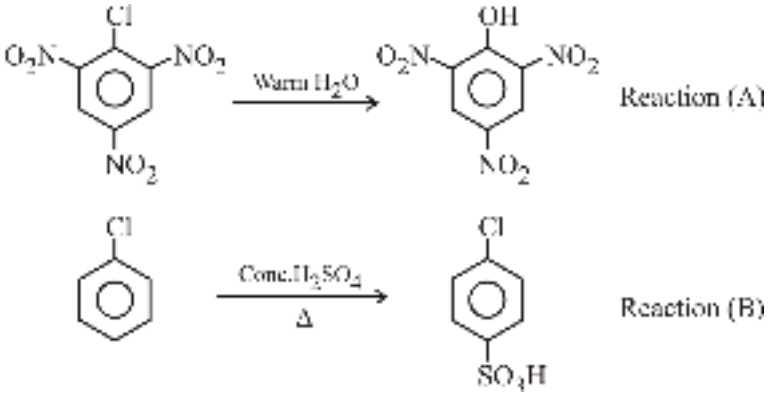
20) **Assertion (A) :-** 1-Propanol have higher boiling point than methoxyethane.
**Reason (R) :-** Both have intermolecular hydrogen bonding but 1-propanol have strong hydrogen bonding.

(1) Both assertion and reason are true and reason is correct explanation of assertion.

(2) Both assertion and reason are true but reason is not correct explanation of assertion

(3) Assertion is true but reason is false.

(4) Assertion and reason both are false.




21) Consider following reaction :- Find correct statement :-

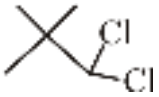
(1) Both reaction are example of substitution reaction

(2) Reaction A is electrophilic substitution reaction

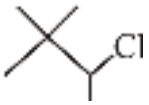
(3) Reaction B is nucleophilic substitution reaction

(4) All are correct

22) Which of the following is 1-Chloro-2,2-dimethylpropane :-

(1) 


(2) 


(3) 


(4) 


23) Which of the following will gives Lassaigne's test :-

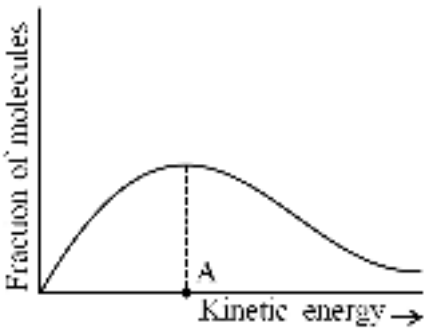
(1) $CH_3–CH_2–NH_2$

(2) $CH_3–CH_2–Cl$

(3) $CH_3–CH_2–SH$

(4) All of these

24) In the following curve, point A represent :-



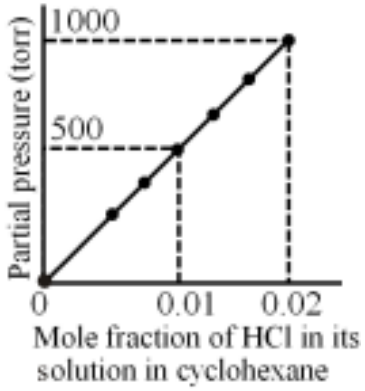

(1) Energy of activation

(2) Most probable kinetic energy

(3) Energy of activated complex

(4) Total number of molecules

25) If we draw a graph between partial pressure of HCl gas versus mole fraction of the gas in solution, then we get following graph :-




The Henry's law constant, $K_H$ is -

(1) $5 \times 10^4$ torr

(2) $10^4$ torr

(3) $10^{-4}$ torr

(4) $2 \times 10^{-5}$ torr

26) **Assertion (A) :** It is possible to calculate $\wedge_m^o$ of electrolyte from the $\wedge_m^o$ of individual ions only for strong electrolyte.
**Reason (R) :** For weak electrolytes, it is possible to determine the value of it's dissociation constant once we know the $\wedge_m^o$ and $\wedge_m$ at a given concentration C.

(1) Both Assertion and Reason are correct and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are correct but Reason is not correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

27) The pair of compounds that can work both as an oxidising as well as reducing agent is :
(i) $SO_2$ (ii) $H_3PO_3$ (iii) $NH_3$ (iv) $HNO_3$

(1) (i) and (iii)

(2) (ii) and (iii)

(3) (i) and (ii)

(4) (iii) and (iv)

28) The maximum number of atoms are present in :-

(1) 5.6 L of $CO_2$(g) at STP

(2) 0.5 g of $H_2$(g)

(3) 18 mL of water (d = 1 g/m□)

(4) 16 g of $O_3$

29) In which of the following reaction hybridisation of central atom does not change :-

(1) $PCl_5 + 5H_2O \xrightarrow{\Delta}$

(2) $SF_4 + 4H_2O \xrightarrow{\Delta}$

(3) $XeF_6 + 6H_2O \xrightarrow{\Delta}$

(4) $SiCl_4 + 4H_2O \xrightarrow{\Delta}$

30) Match the column :-

| | **Chemical Reaction** | | **Property of product** |
|---|---|---|---|
| (a) | $CuSO_4 + NH_4OH$ (excess) → | (P) | Square planar |
| (b) | $HgCl_2 + KI$ (excess) → | (Q) | $sp^3$ hybridisation |
| (c) | $ZnSO_4 + NH_4OH$ (excess) → | (R) | Diamagnetic |
| (d) | $NiSO_4$ + dmg → | (S) | Dark red ppt |

(1) a-P ; b-R ; c-S ; d-P,R

(2) a-R,S ; b-P,R ; c-R,S ; d-Q,S

(3) a-P ; b-Q,R ; c-Q,R ; d-P,R,S

(4) a-R,R ; b-Q,R ; c-Q,R ; d-P,R

31) Which of the following is not a chiral compound :-

(1) $[Co(NH_3)_3Cl_3]$

(2) $[CoCl_2(OX)_2]^{3-}$

(3) $[Fe(en)(NH_3)_2Cl_2]$

(4) $[Cr(gly)_3]$

32) In which of the following arrangement the order is not according to the property indicate against it -

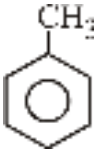
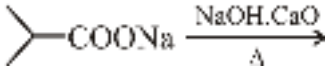
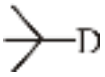
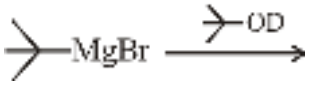
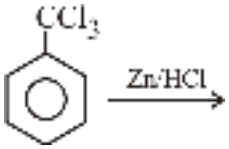
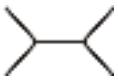
(1) Sc < Y < La (Atomic size)

(2) C < B < N < O ($2^{nd}$ Ionisation enthalpy)

(3) I < Br < F < Cl (Increasing electron gain enthalpy with negative sign)

(4) B > Al > Ga > In > Tl (Electronegativity)

33)

| | | | |
|---|---|---|---|
| a. |  | P. |  |
| b. |  | Q. |  |
| c. |  | R. |  |
| d. |  | S. |  |

(1) a-S, b-R, c-Q, d-P

(2) a-R, b-S, c-Q, d-P

(3) a-S, b-R, c-P, d-Q

(4) a-P, b-Q, c-R, d-S

34) The heat of combustion of $C_4H_{10}$ is –2878 kJ $mol^{-1}$. If the heats of formation of $CO_2$ and $H_2O$ are –393.5 kJ $mol^{-1}$ and –285.8 kJ $mol^{-1}$ then the heat of formation of $C_4H_{10}$ is :-

(1) –125.0 kJ $mol^{-1}$

(2) 126.75 kJ $mol^{-1}$

(3) –402.5 kJ $mol^{-1}$

(4) 402.5 kJ $mol^{-1}$

35) The rate for a first-order reaction is $0.6932 \times 10^{-2}$ mol $L^{-1}$ $min^{-1}$ and the initial concentration of the reactants is 1M, $t_{1/2}$ is equal to :-

(1) 6.932 min

(2) 100 min

(3) $0.6932 \times 10^{-3}$ min

(4) $0.6932 \times 10^{-2}$ min

36) Which of the following is correct for the reaction $3X \rightarrow X_3$ ($\alpha$ represents degree of association)

(1) $\alpha = \frac{3}{2}(1-i)$

(2) $\alpha = \frac{2}{3}(1-i)$

(3) $\alpha = 2 \times 3\,(1-i)$

(4) $\alpha = 3\,(1-i)$

37) P molal solution of a compound in benzene has mole fraction of solute equal to 0.2. The value of "P" is

(1) 2.0

(2) 13.88

(3) 14.3

(4) 3.2

38) Which reaction of Xenon compound is not feasible?

(1) $3XeF_4 + 6H_2O \rightarrow 2Xe + XeO_3 + 12HF + \frac{3}{2} O_2$

(2) $2XeF_2 + 2H_2O \rightarrow 2Xe + HF + O_2$

(3) $XeF_6 + RbF \rightarrow Rb\,[XeF_7]$

(4) $XeO_3 + 6HF \rightarrow XeF_6 + 3H_2O$

39) Which of the following does not follow EAN rule ?

(1) $[Ni(CO)_4]$

(2) $[Fe(CO)_5]$

(3) $[Cr(CO)_6]$

(4) $[Mn(CO)_6]$

40) Which of the following has β-Glycosidic linkage.

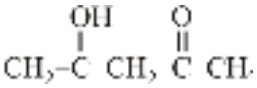
(1) Maltose

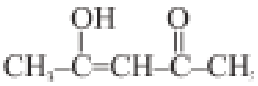
(2) Lactose

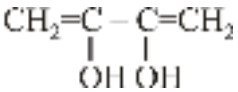
(3) Glycogen

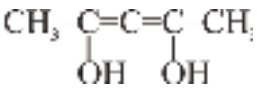
(4) Starch

41) The most stable enolic form of 2, 4–pentanedione is :-

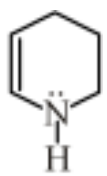
(1) 

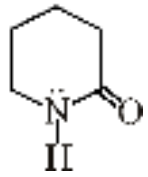
(2) 

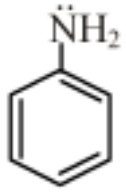
(3) 

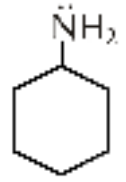
(4) 

42) Which of the following is most basic?

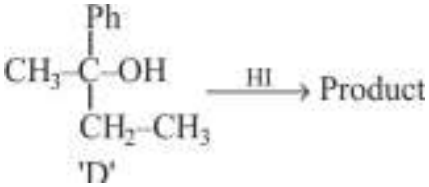
(1) 

(2) 

(3) 

(4) 

43) The following reaction is suppose to take place through $S_N1$ mechanism




'D'

If the configuration of substrate is D, then configuration of products will be :-

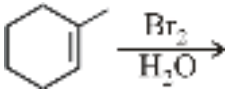
(1) D

(2) L

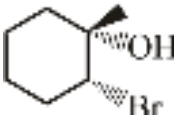
(3) 50% D & 50% L

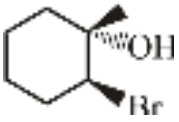
(4) May be D or L

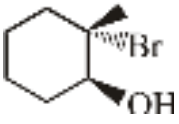
44) Which is correct major product of this reaction?

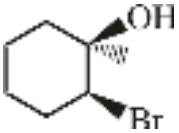



Major product

(1) 

(2) 

(3) 

(4) 

45) A colourless sol. (A) gives black ppt. on passing $H_2S$. (A) also gives a white ppt. with $SnCl_2$ which gradually changes to grey. Identify (A) :-

(1) $ZnCl_2$

(2) $CdBr_2$

(3) $HgCl_2$

(4) $Cu(NO_3)_2$

## BIOLOGY

1) The body is externally and internally divided into segments with a serial repetition of at least some organs in

(1) Hydra

(2) Fasciola

(3) Earthworm

(4) Pila



2) Which of the following character is not true for given figure :

(1) Fertilisation is internal

(2) Flame cells help in osmoregulation and excretion

(3) Development is through many larval stages

(4) Sexes are seperate

3) Which of the following statements is not correct ?

(1) Cyclostomes are marine but migrate for spawning to fresh water.

(2) Over one-third of all named species on Earth are Arthropods.

(3) Platyhelminthes are mostly endoparasites found in animals including human beings.

(4) Respiration in Amphibians is by gills, lungs & skin.

4) How many animals in the list given below have radial symmetry :
*Obelia*, Sponge, *Aurelia*, *Pleurobranchia*, *Fasciola*, *Ascaris*, Earthworm, *Pila*, Larva of star fish.

(1) Two

(2) Three

(3) Five

(4) Four

5) Multilayered epithelium is not found in -

(1) Pharynx

(2) Moist surface of buccal cavity

(3) Pancreatic duct

(4) Lining of body cavity

6) Which part gets reduced in the alimentary canal of frog due to its carnivorous nature :

(1) Oesophagus

(2) Intestine

(3) Buccal cavity

(4) Cloaca

7) Identify X, Y & Z

| **Partial pressure** | **Alveoli** | **Deoxygenated Blood** | **Oxygenated Blood** | **Tissue** |
|---|---|---|---|---|
| $pO_2$ | 'X' | 40 | 95 | 'Z' |
| $pCO_2$ | 40 | 'Y' | 40 | 45 |

| | **X** | **Y** | **Z** |
|---|---|---|---|
| (1) | 104 | 40 | 45 |
| (2) | 45 | 40 | 104 |
| (3) | 40 | 40 | 104 |
| (4) | 104 | 45 | 40 |

(1) 1

(2) 2

(3) 3

(4) 4

8) Lungs are covered by a double layered membrane which is called

(1) Pleural membrane

(2) Visceral membrane

(3) Pulmonary membrane

(4) Parietal membrane

9) **Statement-I :** Fibres of the tracts are covered with the myelin sheath, which constitute the outer part of cerebral hemisphere.
**Statement-II :** The hypothalamus contains a number of centres which control body temperature, respiration and cardiovascular reflexes.
In the light of the above statements, choose the most appropriate answer from the options given below :-

(1) Both statement-I and statement-II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Both statement-I and statement-II are incorrect.

(4) Statement-I is incorrect but statement-II is correct.

10) Which of the following movements restores the resting potential of the membrane at the site of excitation ?

(1) $Na^+$ diffuses outside

(2) $K^+$ diffuses inside

(3) $K^+$ diffuses outside

(4) $Na^+$ diffuses inside

11) Read the following statements (a-d) :-
(a) Hormones are non-nutrient chemicals which act as intercellular messengers and are produced in trace amounts.
(b) The endocrine system alone co-ordinate and regulate the physiological functions in the body
(c) Hypothalamus forms the basal part of diencephalon
(d) Somatostatin from hypothalamus inhibits release of growth hormone from pituitary
How many of the above statement are true ?

(1) Two

(2) One

(3) Four

(4) Three

12) Which statement is incorrect.

(1) Adrenal cortex hormones stimulates $Na^+$ reabsorption.

(2) Thyroid hormone helps in regulation of menstrual cycle.

(3) Estradiol and oestrogen bind with intranuclear receptors.

(4) Cortisol maintain immune system by stimulation of Antibody formation.

13) All of the following are gastrointestinal hormones except -

(1) Gastrin

(2) Secretin

(3) Cholecystokinin

(4) Erythropoitin

14) Match the excretory organ listed under column I with the animals given under column II choose the answer which gives correct combination of alphabets of the two column.

| **Column I (Excretory organ)** | | **Column II (Animals)** | |
|---|---|---|---|
| (A) | Nephridia | (P) | Hydra |

| | | | |
|---|---|---|---|
| (B) | Malpighian tubules | (Q) | Earthworm |
| (C) | Protonephridia | (R) | Shark |
| (D) | Kidney | (S) | Platyhelminthes |
| | | (T) | Cockroach |

(1) A = Q, B = T, C = S, D = R

(2) A = S, B = Q, C = P, D = T

(3) A = T, B = Q, C = S, D = R

(4) A = Q, B = S, C = T, D = P

15) Select the structures which form the renal tubule of a nephron.

(1) Bowman's capsule + Renal corpuscle

(2) PCT + DCT + Glomerulus

(3) Bowman's capsule + PCT + DCT + Loop of Henle

(4) PCT + DCT + Loop of Henle + Collecting Duct

16)

Muscles with characteristic striations and involuntary are

(1) muscles in the wall of alimentary canal

(2) muscles of the heart

(3) muscles assisting locomotion

(4) muscles of the eyelids

17) Choose the correct sequence of vertebrae.

(1) Cervical → Lumbar → Thoracic → Sacral → Coccyx

(2) Cervical → Thoracic → Lumbar → Coccyx → Sacral

(3) Cervical → Sacral → Lumbar → Thoracic → Coccyx

(4) Cervical → Thoracic → Lumbar → Sacral → Coccyx

18) **Assertion :** The male sex accessory ducts include rete testis, vasa efferentia, epididymis and vas deferens.
**Reason :** The epididymis leads to vas-deferens that ascends to the abdomen and loops over the urinary bladder.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

19) As the ventricle relaxes, the intra ventricular pressure ____. When ventricular pressure drops below atrial pressure the ____ opens, and ventricular filling begins :-

(1) Rises, Bicuspid valves

(2) Falls, Semilunar valves

(3) Rises, Mitral valves

(4) Falls, AV valves

20) **Statement-I :** In atherosclerosis only deposition of fat and fibrous tissues occurs, which make the wall of arteries hard.
**Statement-II :** In heart failure congestion of the liver is only main region.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

21) How many secondary spermatocytes will form 100 sperm ?

(1) 25

(2) 50

(3) 100

(4) 200

22) Which of the following statement is incorrect ?

(1) The process of formation of a mature female gamete is called oogenesis.

(2) Oogenesis is initiated just after the birth of girl child.

(3) Tertiary follicle is characterised by a fluid filled cavity called antrum.

(4) The secondary oocyte retain bulk of the nutrient rich cytoplasm of the primary oocyte

23) In the given diagramatic structures, which one will be embedded in the endometrium of uterus ?

(1) 

(2) 

(3) 

(4) 

24) Which one of the following is the most widely accepeted method of contraceptive in India, as at

present ?

(1) Cervical cap
(2) Vasectomy
(3) MALA-D
(4) IUD

25) Match the column-A with column-B.

| Column-A | | Column-B | |
|---|---|---|---|
| A | Menstrual phase | i | Corpus luteum |
| B | Follicular phase | ii | Breakdown of endometrium |
| C | Ovulatory phase | iii | Fully mature graafian follicle |
| D | Luteal phase | iv | Release of ovum |

(1) A-iv, B-ii, C-i, D-ii
(2) A-ii, B-i, C-iv, D-iii
(3) A-ii, B-iii, C-iv, D-i
(4) A-iv, B-i, C-ii, D-iii

26) In a family husband with A blood group and wife with O blood group then, which of following blood group is not possible in progenies ?

(1) A
(2) B only
(3) O
(4) Both B and AB

27) **Statement-I :** Based upon observations on dihybrid crosses, Mendel proposed the law of independent assortment.
**Statement-II :** In $F_2$ generation of dihybrid cross Mendel obtained 75% plants having round seed.

(1) Both statement I and II are incorrect.
(2) Statement I is correct but statement II is incorrect.
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and II are correct.

28) Good humor hypothesis of health was given by ?

(1) Hippocrates
(2) Louis pasteur
(3) William Harvey
(4) Edward Jenner

29) **Assertion (A):** Tumour cells have been shown to avoid detection & destruction by immune system.

**Reason (R):** α-interferons are given which activate immune system & help in destroying the tumour.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is false but (R) is true

(4) (A) is true but (R) is false

30) Tasmanian wolf and tiger cat are example of :

(1) adaptive radiation

(2) convergent evolution

(3) adaptive convergence

(4) mutation

31) Match the following terms with their definitions :

| Term | | Definition | |
|---|---|---|---|
| i. | Homology | A. | Different structures evolving for the same function |
| ii. | Analogy | B. | Same structure evolving in different species for different functions. |
| iii. | Divergent evolution | C. | Selection of similar adaptive features in different organisms. |
| iv. | Convergent evolution | D. | Structures derived from a common ancestor evolving differently |

(1) i-A, ii-B, iii-C, iv-D

(2) i-C, ii-A, iii-B, iv-D

(3) i-D, ii-A, iii-B, iv-C

(4) i-D, ii-B, iii-C, iv-A

32) Contraceptive oral pills help in birth control by :

(1) Killing the sperm in uterus

(2) Forming barriers between sperms and ova.

(3) Preventing ovulation

(4) Killing the ova

33) Mycorrhizae are example of mutualism. The fungi help the plant in the absorption of essential _____ from the soil while the plant in turn provides the fungi with energy - yielding _____.

(1) Carbohydrate, Chemical

(2) Nutrients, Nucleic acid

(3) Carbohydrate, Nutrients

(4) Nutrients, Carbohydrates

34) Match the Column-I and Column-II choose the correct options given.

| Column-I | | Column-II | |
|---|---|---|---|
| a | Primary Producer | i | Grasshopper |
| b | Primary Consumer | ii | Wolf |
| c | Secondary Consumer | iii | Lion |
| d | Tertiary Consumer | iv | Grass |

(1) a - i, b- iii, c - ii, d - iv

(2) a - ii, b- iii, c - iv, d - i

(3) a - iv, b- i, c - ii, d - iii

(4) a - i, b- iv, c - ii, d - iii

35) The IUCN Red list (2004) documents the extinction of __A__ species (including __B__ vertebrates, __C__ invertebrates and __D__ plants) in the last 500 years.

(1) A=784, B=338, C=359, D=87

(2) A=338, B=784, C=87, D=359

(3) A=784, B=359, C=338, D=87

(4) A=784, B=338, C=359, D=187

36) **Statement 1 :** $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$ represents logistic growth.
**Statement 2 :** Since resources for growth for most animal populations are finite and become limiting sooner or later, the logistic growth model is considered a more realistic one.
In the light of above statements, choose the correct answer from the options given below.

(1) Statement 1 is correct and Statement 2 is incorrect.

(2) Statement 1 is incorrect and Statement 2 is correct.

(3) Both Statement 1 and Statement 2 are correct.

(4) Both Statement 1 and Statement 2 are incorrect.

37) Match the following Column-I with Column-II :

| Column-I | | Column-II | |
|---|---|---|---|
| (A) | Detritus food chain | (i) | Detritivores break down detritus into smaller particles |
| (B) | Grazing food chain | (ii) | Bacterial and fungal enzymes degrade detritus into simpler inorganic substance |
| (C) | Fragmentation | (iii) | Major conduit for energy flow in aquatic ecosystem |

| | | | |
|---|---|---|---|
| (D) | Catabolism | (iv) | Major conduit for energy flow in terrestrial ecosystem |

(1) A - (iv), B - (iii), C - (i), D - (ii)

(2) A - (iv), B - (iii), C - (ii), D - (i)

(3) A - (iii), B - (iv), C - (i), D - (ii)

(4) A - (iii), B - (iv), C - (ii), D - (i)

38) Number of chromatids in chromosome at metaphase is :-

(1) Two each in mitosis and meiosis

(2) Two in mitosis and one in meiosis

(3) Two in mitosis and four in meiosis

(4) One in mitosis and two in meiosis

39) Given below are two statements :-
**Statement-I :** The cis and trans faces of the Golgi body are entirely different but interconnected.
**Statement-II :** Golgi apparatus is the important site of glycolipid and glycoprotein synthesis.
In the light of the above statements, choose the most appropriate answer from option given below :

(1) Statement-I and statement-II are incorrect

(2) Statement-I is correct and statement-II is incorrect

(3) Statement-I and statement-II both are correct

(4) Statement-I is incorrect and statement-II is correct

40) Match **List-I** with **List-II**.

| | **List-I** | | **List-II** |
|---|---|---|---|
| A. | Cell are metabolically active and proliferate | I. | $G_2$ phase |
| B. | DNA replication takes place | II. | $G_1$ phase |
| C. | Proteins are synthesised | III. | $G_0$ phase |
| D. | Quiescent state with metabolically active cells | IV. | S phase |

Choose the **correct** answer from the options given below :

(1) A-IV, B-II, C-III, D-I

(2) A-I, B-III, C-IV,D -II

(3) A-II, B-I, C-III, D-IV

(4) A-II, B-IV, C-I, D-III

41) Morphology of chromosomes is most easily studied in :

(1) Prophase

(2) Metaphase

(3) Telophase

(4) Interphase

42) Given below are two statement: One is labelled as assertion A and the other is labelled as reason R.
**Assertion (A) :** Ecosystem may be large or small
**Reason (R) :** Ecosystem varies greatly in size from a small pond to large forest or sea.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both A and R are true but R is not the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true and R is the correct explanation of A.

43) Population ecology is an important area of ecology because it links ecology to _____.

(1) Population density

(2) Population growth

(3) Population genetics and evolution

(4) Population interactions

44) (A) Tropics harbour more species than temperate or polar areas.
(B) Colombia located near the equator has nearly 1400 species of birds
(C) New York at 41°N has 105 species and Greenland at 71°N only 56 species.
(D) A forest in a tropical region like Equador has up to 10 times as many species of vascular plants as a forest equal area in a temperate region like the midwest of the USA.
Choose the correct statements from the option given below :-

(1) A, B, C, D

(2) A and B

(3) A, B, D

(4) A, C and D

45) **Statement-I :** Habitat loss and fragmentation is the most important cause driving animals and plants to extinction.
**Statement-II :** When a species become extinct the plant and animal species associated with it in an obligatory way also become extinct.

(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect.

(3) Only Statement-I is correct.

(4) Only Statement-II is correct.

46) Match the column A and Column B and chose correct option :

| | **Column – A** | | **Column - B** |
|---|---|---|---|
| (A) | Cardiac glycosides | (i) | Acacia, Cactus |

| | | | |
|---|---|---|---|
| (B) | Distasteful due to a special chemical | (ii) | Insects |
| (C) | Thorns | (iii) | *Calotropis* |
| (D) | Camouflage | (iv) | Monarch butterfly |

(1) A-(iii), B-(iv), C-(i), D-(ii)

(2) A-(i), B-(ii), C-(iii), D-(iv)

(3) A-(iv), B-(iii), C-(ii), D-(i)

(4) A-(iii), B-(iv), C-(ii), D-(i)

47) Which of the following you will not prefer for the formation of rDNA ?

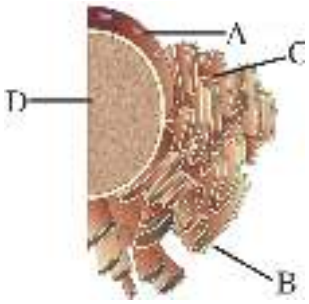
(1) Choice of vector whose origin support high copy number

(2) Use of same restriction enzyme for cleaving vector and desired gene

(3) Bacterial cells must be made competent to take up the DNA

(4) Presence of more than one restriction sites of a restriction enzyme

48) In the following diagram, select the part **correctly** matched with its character or function :-




(1) A → The passage through which movement of RNA and protein molecules take place in both direction

(2) B → Membrane bound structure composed of RNA and protein

(3) C → Frequently observed in cells actively involved in lipid synthesis

(4) D → Provide surface for DNA and protein synthesis

49) Match the column-I with column-II & give the answer.

| **Column-I (Common Name)** | | **Column-II (Order)** | |
|---|---|---|---|
| (I) | Man | (a) | Diptera |
| (II) | House fly | (b) | Poales |
| (III) | Wheat | (c) | Primata |
| (IV) | Mango | (d) | Sapindales |

(1) (I)-(a), (II)-(b), (III)-(c), (IV)-(d)

(2) (I)-(c), (II)-(a), (III)-(b), (IV)-(d)

(3) (I)-(c), (II)-(b), (III)-(a), (IV)-(d)

(4) (I)-(a), (II)-(b), (III)-(d), (IV)-(c)

50) Match the column-I with column-II and select correct option :

| Column-I | | Column-II | |
|---|---|---|---|
| A | Dinoflagellates | I | Plasmodium |
| B | Diatoms | II | Longitudinal and transverse flagella |
| C | Euglenoids | III | Presence of silica in cell wall |
| D | Slime moulds | IV | Pellicle |

(1) A – II, B – III, C – IV, D – I

(2) A – III, B – II, C – IV, D – I

(3) A – IV, B – III, C – II, D – I

(4) A – I, B – II, C – III, D – IV

51) **Assertion (A) :** In gymnosperms the female gametophytes do not have an independent free living existance.
**Reason (R) :** In gymnosperms female gametophytes remain within the sporangia retained on sporophyte.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) (A) is correct but (R) is incorrect.

(3) (A) is incorrect but (R) is correct.

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

52) Find the incorrect statement

(1) Light saturation for $CO_2$ fixation occurs at 10% of full sunlight

(2) PEP is the primary $CO_2$ acceptor in $C_4$ plants

(3) Z-scheme of light reaction takes place in presence of PS-I only

(4) Grana lamellae have both PS-I and PS-II

53) In $C_4$ plants malic acid or aspartic acid (4C) formed in ___(a)___ cells are transported to ___(b)___ cells.

(1) (a) mesophyll (b) bundle sheath

(2) (a) bundle sheath (b) mesophyll

(3) (a) mesophyll (b) mesophyll

(4) (a) bundle sheath (b) bundle sheath

54) In non-cyclic photophosphorylation, which of the following is not synthesised.

(1) ATP

(2) $NADPH + H^+$

(3) $O_2$

(4) Starch

55) **Assertion :-** Cytochrome c is a small protein attached to the outer surface of the inner mitochondrial membrane.
**Reason :-** Cytochrome c act as a mobile carrier for transfer of electrons between complex - III and IV.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

56) Which enzyme is used to convert pyruvic acid to lactic acid ?

(1) Pyruvic acid Decarboxylase
(2) Lactate Dehydrogenase
(3) Alcohol Dehydrogenase
(4) Pyruvate Dehydrogenase

57) Read the following statements carefully.
(I) Ethephon hastens fruit ripening in tomatoes and apples.
(II) Ethylene is one of the most widely used PGR in agriculture.
(III) Kinetin does not occurs naturally in plants.
(IV) 2,4-D, widely used to kill monocotyledonous weeds, does not affect mature dicotyledonous plants.
(V) ABA plays an important role in seed development, maturation and dormancy.
(VI) Auxin was isolated by E. Kurosawa from tips of coleoptiles of oat dormancy.

(1) only I,II,III are correct
(2) only I,II,IV,V are correct
(3) only IV,VI are incorrect
(4) only III,IV,V are incorrect

58) **Assertion (A) :** In plants such as bean, gram, pea, the endosperm is not present in mature seeds.
**Reason (R) :** Endosperm is consumed during embryo development in plants such as bean, gram, pea.

(1) Both (A) and (R) are correct and (R) is the correct explaination for (A)
(2) (A) is correct but (R) is incorrect
(3) (A) is incorrect but (R) is correct
(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

59) Given floral formula is related to which plant :- $\oplus \male\!\female\, K_{(5)} \overbrace{C_{(5)} A}_{5}\ \underline{G}_{(2)}$

(1) Makoi
(2) Onion
(3) Pea

(4) Mustard

60) Arrange the following layers in young dicot stem from outside to inside :
(A) Hypodermis
(B) Cortex
(C) Pericycle
(D) Endodermis
(E) Pith

(1) D→B→C→E→A

(2) A→B→D→C→E

(3) A→B→C→D→E

(4) D→B→A→E→C

61) The hypodermis of monocot stem consist of :-

(1) Parenchymatous cells

(2) Collenchymatous cells

(3) Meristmetic cells

(4) Sclerenchymatous cells

62) Pollen grains are generally .....measuring about .....in diameter.

(1) Spherical, 25-50 micrometers

(2) Rectangular, 50-75 micrometers

(3) Triangular, 25-50 micrometers

(4) Disc shaped , 50-60 micrometers

63) Read the following statements :
A. Endosperm development precedes embryo development.
B. The coconut water from tender coconut is nothing but free nuclear endosperm.
C. Early stages of embryo development are different in monocotyledons and dicotyledons.
D. In grass family the cotyledon is called scutellum.
How many of the above statement is/are **correct**.

(1) Three

(2) Four

(3) One

(4) Two

64) (i) Promote nutrient mobilisation.
(ii) Overcome apical dominance.
(iii) Delay in leaf senescence.
(iv) Adventitious shoot formation.
All the above physiological effects are related to which of the following plant growth regulators ?

(1) ABA

(2) IBA

(3) $GA_3$

(4) Cytokinins

65) The _____ is a scar on the seed coat through which the developing seeds were attached to the fruit

(1) Micropyle

(2) Hilum

(3) Testa

(4) Tegmen

66) **Statement-I :** In brown algae food is stored as floridean starch which is very similar to amylopectin and glycogen in structure.
**Statement-II :** The members of phaeophyceae are found primarily in marine habitats.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

67) Cr-Jacob disease (CJD) in humans is caused by :-

(1) Viroids

(2) Virus

(3) Prions

(4) Lichens

68) Which of the following protein helps in the transport of glucose across cell membrane ?

(1) Trypsin

(2) Collagen

(3) GLUT-4

(4) Antibody

69) Read the following statements (A-D)-
**(A)** Classification is the process by which anything is grouped into convenient categories based on some easily observable, characters.
**(B)** Based on characteristics, all living organisms can be classified into same taxa.
**(C)** Earliest classification were based on the uses of various organisms.
**(D)** Characterisation, identification, classification and nomenclature are the process that are basic to taxonomy.
How many to the above statements are correct?

(1) One

(2) Two

(3) Four

(4) Three

70) **Statement-I :** Besides the involvement of multiple genes polygenic inhertance also takes into account the influence of environment.
**Statement-II :** Mendelian disorders are caused due to absence or excess or abnormal arrangement of one or more chromosomes.

(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct and Statement-II are incorrect.
(3) Statement-I is incorrect and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

71) Which codon has dual functions ?

(1) UUU
(2) UUC
(3) UGG
(4) AUG

72) **Assertion :-** The total estimated cost of the human genome project was approximately 9 billion US dollars.
**Reason :-** Human genome is said to have approximately $3 \times 10^9$ bps and cost of sequencing was 3 US dollar per bp.

(1) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion
(2) Assertion is true but the Reason is false
(3) Both Assertion and Reason are true and Reason is a correct explanation of the Assertion
(4) Both Assertion and Reason is false

73) Read the given following statements:
(a) It is now proven that DNA replicates semi conservatively.
(b) Heavy DNA molecule ($N^{15}$) could be distinguished from the normal DNA ($N^{14}$) by centrifugation in a cesium chloride (CsCl) density gradient.
(c) In living cells, such as *E.coli*, the process of replication requires a set of catalysts (enzymes).
(d) In *E.coli*, the process of DNA replication completes within 28 minutes.
(e) Any mistake during replication would result into mutations.
Choose the correct statements from the options given below:

(1) (b), (d), (e) only
(2) (a), (b), (c), (e)
(3) (a), (c), (e) only
(4) (a), (c) only

74) When pure round and yellow seeded pea plant (RRYY) is crossed with wrinkled and green seed pea plant (rryy), and the $F_1$ progeny is test crossed then what phenotypic ratio will be obtained :-

(1) 9 : 3 : 3 : 1
(2) 1 : 2 : 1 : 2 : 4 : 2 : 1 2 : 1
(3) 1 : 1

(4) 1 : 1 : 1 : 1

75) Which of the following is not correct about translation ?

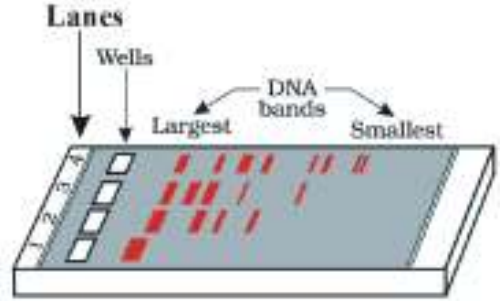
(1) It start with AUG

(2) Stops at termination codon

(3) Occur in cytoplasm

(4) Occur in nucleus

76) In following figure of gel electrophoresis which lane is showing undigested set of DNA sample ?




(1) lane - 1

(2) lane - 2

(3) lane - 3

(4) lane - 4

77) Match the column-I and column-II

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (i) | Sutton and Boveri | (a) | Linkage map |
| (ii) | Morgan | (b) | Linkage |
| (iii) | Alfred Sturtevant | (c) | X-body |
| (iv) | Henking | (d) | Chromosomal theory of inheritance |

(1) i-d, ii-b, iii-a, iv-c

(2) i-d, ii-a, iii-b, iv-c

(3) i-a, ii-b, iii-c, iv-d

(4) i-b, ii-c, iii-d, iv-a

78) **Assertion (A) :** Sutton and Boveri noted that the behaviour of chromosomes was parallel to the behaviour of genes.
**Reason (R) :** Frequency of recombination between gene pairs on same chromosomes is used as measure of distance between genes.

(1) A and R both correct but R is not correct explanation of A.

(2) A and R both correct and R is correct explanation of A.

(3) A is correct and R is incorrect.

(4) A is incorrect and R is correct.

79) As per ABO blood grouping system, the genotype of father is $I^A I^B$, mother is $I^B$ i. Genotype of child can be :-

(1) $I^A I^B$ / $I^A$ i / $I^B$ i / $I^B I^B$

(2) $I^A I^B$ / $I^A$ i / $I^B$ i / i i

(3) $I^A I^B$ / $I^A I^A$ / $I^B$ i / $I^B I^B$

(4) $I^A I^B$ / $I^A I^A$ / i i / $I^B I^B$

80) Choose correct statements from given below :-
(a) RNA being a catalyst was reactive and stable.
(b) DNA has evolved from RNA with chemical modifications
(c) DNA resists changes by evolving a process of repair.
(d) DNA is more stable than RNA

(1) a, b, c, d

(2) a, c, d

(3) a, b, c

(4) b, c, d

81) **Statement – I :** The two chains of DNA are coiled in a right handed fashion and the pitch of the helix is 34 nm.
**Statement – II :** The plane of one base pair stacks over the other in double helix, this in addition to phosphodiester bonds confers stability to the helical structure.

(1) Statement I is correct and statement II is incorrect

(2) Statement I is incorrect and statement II is correct

(3) Statement I and II both are correct

(4) Statement I and II both are incorrect

82) How many amino acids will be present in a polypeptide chain if chain is coded by given m-RNA ?
5'-GUC ACU GUC GCA AUG GGG CUG UUU CCC ACA UGA CAU -3'

(1) 6 amino acid

(2) 12 amino acid

(3) 11 amino acid

(4) 7 amino acid

83) In pea plant gene of size of starch grain also controls :-

(1) Flower colour

(2) Seed shape

(3) Pod shape

(4) Height

84) Select the genotype amongst given which will result in light skin colour in humans :

(1) AAbbCC

(2) AABBCc
(3) aaBBCC
(4) AabbCc

85) Biological control agent is obtained from :-

(1) *Bacillus thuringiensis*
(2) *E.coli*
(3) *Agrobacterium tumifaciens*
(4) *Meloidogyne incognitia*

86)

Ti plasmid transforms cell of :-

(1) Animals
(2) Plants
(3) Bacteria
(4) Fungi

87)

Silencing of a specific mRNA due to a complementary dsRNA molecule is done by inhibiting which process?

(1) Replication
(2) Transcription
(3) Reverse transcription
(4) Translation

88) **Statement-A :** At present about 30 recombinant therapeutics have been approved for human use the world over.
**Statement-B :** 27 documented varieties of Basmati rice are grown in India.

(1) Both A and B are true
(2) Only A is true
(3) Only B is true
(4) Both A and B are false

89) Which is the common microbe used for both bread making and wine making ?

(1) *Saccharomyces cerevisiae*
(2) *Monascus purpureus*
(3) *Cyclosporin*
(4) All of the above

90) Which of the following peptides are present in mature insulin?

(1) A-peptide and B-peptide
(2) A-peptide and C-peptide
(3) B-peptide and C-peptide
(4) Only C-Peptide

ANSWER KEYS

## PHYSICS

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 2 | 1 | 1 | 4 | 2 | 1 | 1 | 4 | 4 | 2 | 1 | 3 | 2 | 3 | 3 | 3 | 1 | 3 | 4 | 4 |
| Q. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| A. | 3 | 4 | 2 | 3 | 1 | 1 | 2 | 2 | 2 | 1 | 2 | 2 | 4 | 4 | 1 | 2 | 3 | 2 | 2 | 3 |
| Q. | 41 | 42 | 43 | 44 | 45 | | | | | | | | | | | | | | | |
| A. | 3 | 3 | 4 | 4 | 2 | | | | | | | | | | | | | | | |

## CHEMISTRY

| Q. | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 4 | 1 | 3 | 4 | 3 | 2 | 2 | 4 | 4 | 3 | 3 | 2 | 2 | 1 | 4 | 4 | 2 | 3 | 2 | 3 |
| Q. | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |
| A. | 1 | 2 | 4 | 2 | 1 | 4 | 3 | 3 | 4 | 3 | 1 | 4 | 1 | 1 | 2 | 1 | 4 | 4 | 4 | 2 |
| Q. | 86 | 87 | 88 | 89 | 90 | | | | | | | | | | | | | | | |
| A. | 2 | 4 | 3 | 2 | 3 | | | | | | | | | | | | | | | |

## BIOLOGY

| Q. | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 3 | 4 | 2 | 2 | 4 | 2 | 4 | 1 | 3 | 3 | 4 | 4 | 4 | 1 | 3 | 2 | 4 | 2 | 4 | 4 |
| Q. | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 |
| A. | 2 | 2 | 3 | 4 | 3 | 4 | 4 | 1 | 2 | 1 | 3 | 3 | 4 | 3 | 1 | 3 | 1 | 1 | 3 | 4 |
| Q. | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 |
| A. | 2 | 4 | 3 | 1 | 1 | 1 | 4 | 1 | 2 | 1 | 1 | 3 | 1 | 4 | 2 | 2 | 3 | 1 | 1 | 2 |
| Q. | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 |
| A. | 4 | 1 | 1 | 4 | 2 | 4 | 3 | 3 | 4 | 2 | 4 | 3 | 2 | 4 | 4 | 1 | 1 | 1 | 1 | 4 |
| Q. | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 | | | | | | | | | | |
| A. | 4 | 1 | 2 | 4 | 1 | 2 | 4 | 1 | 1 | 1 | | | | | | | | | | |

SOLUTIONS

PHYSICS

1)

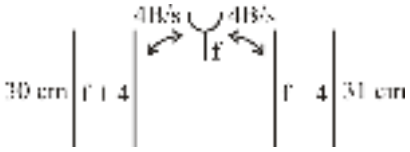
$[E] = [M]^x [L]^y [f]^z = [M]^x [L]^y [T^{-1}]^z$
$[M^1L^2T^{-2}] = [M^x] [L^y] [T^{-z}]$
$x = 1, y = 2, z = 2$
$\Rightarrow x + y + z = 5$

2)

In addition and subtraction, see minimum decimal point.

3) $T = \mu v^2 = 1.3 \times 10^{-4} \times (30)^2 \simeq 0.12$

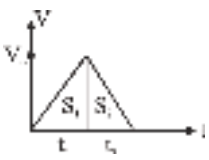
4)



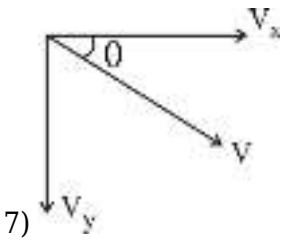

$f \propto \frac{1}{\ell}$
$\frac{f+4}{f-4} = \frac{31}{30}$
$\Rightarrow 30f + 120 = 31f - 124$
$f = 244$ Hz

5) $mg = Kx_0$
$\frac{m}{K} = \frac{x_0}{g} = \frac{0.2}{10}$
$T = 2\pi\sqrt{\frac{m}{K}} = 2\pi\sqrt{\frac{0.2}{10}} = 2\sqrt{\frac{2}{10}}$
$\left\{T = \frac{2}{\sqrt{5}}\text{ sec}\right\}$

6)




$\frac{V_{avg.1}}{V_{avg_2}} = \frac{S_1 t_1}{S_2 t_2} = \frac{\frac{1}{2}V_0 t_1}{t_1}\frac{t_2}{\frac{1}{2}V_0 t_2} = 1$




7)

$V_y = -gt \Rightarrow V_y = -100$

$V_x = 500$

$$\tan\theta = \frac{V_y}{V_x} = \frac{100}{500} \Rightarrow \theta = \tan^{-1}\frac{1}{5}$$

8) $(V_{rms})_{O_2} = (V_{rms})_{H_2}$

$$\frac{T_{O_2}}{M_{WO_2}} = \frac{T_{H_2}}{M_{WH_2}}$$

$$T_{O_2} = \frac{20 \times 32}{2}$$

$T_{O_2} = 320K$

$T_{O_2} = 47°C$

9) $$\frac{94-86}{2} = K\left(\frac{94+86}{2} - 20\right) \quad ....(i)$$

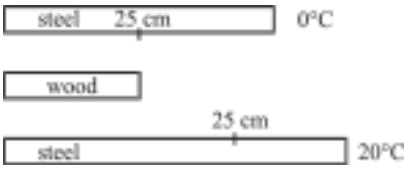
$$\frac{71-69}{t} = K\left(\frac{71+69}{2} - 20\right) \quad ....(ii)$$

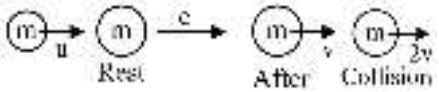
Dividing (i) by (ii), we get

$t = \frac{7}{10}$ min = 42 sec

10) At 0°C reading of scale is 25 cm.

At 20°C, the reading is correct, means the rod length < 25 cm on actual scale (at 20 °C).




11) 


Before collision

by COLM, $m(u) + 0 = m(v) + m(2v)$

$\Rightarrow u = 3v$ .....(1)

$$e = \left|\frac{\vec{v}_2 - \vec{v}_1}{\vec{u}_1 - \vec{u}_2}\right| \Rightarrow e = \frac{2v - v}{u - 0} = \frac{v}{u} = \frac{v}{3v}$$

$$\Rightarrow e = \frac{1}{3}$$

12) $m_1(9 - \Delta r_2) = m_2 \Delta r_2$

or $100(9 - \Delta r_2) = 500\ \Delta r_2$

$$\Delta r_2 = \frac{100 \times 9}{600} = 1.5\,m$$

i.e., Boat moves 1.5 m relative to shore in the direction opposite to the displacement of man.

13) $$\nu = \sqrt{2gh}, t = \sqrt{\frac{2h}{g}}$$

and $$R = 2\sqrt{h(H-h)}$$

At mountain, value of g will be less. Hence, ν will decreases, t will increases and R will remain unchanged.

14) $P_{excess} = P_{in} - P_{out}$

$P_{in} = P_{out} + P_{ex}$

$$P_{in} = P_0 + \rho gh + \frac{2T}{R} \quad ...(1)$$

Now required pressure difference

$\Delta P = P_{in} - P_0$

$$\Delta P = \rho gh + \frac{2T}{R}$$

$$= 1000 \times 10 \times \frac{20}{100} + \frac{2 \times 0.072}{10^{-3}}$$

$= 2144\ N/m^2$

15)

Energy of incident photon of wavelength 2480Å is

$$E = \frac{12400}{2480} eV = 5eV$$

By Einstein's Equation

$K.E._{man} = E - \phi$

$\Rightarrow \phi = E - K.E._{man}$

$= 5eV - 2eV$

$= 3eV$

16) For spherical mirrors

m = +ve ⇒ virtual image

m = –ve ⇒ real image

|m| > 1 ⇒ magnified image

|m| < 1 ⇒ diminished image

17) $$\frac{\phi}{2\pi} = \frac{\Delta x}{\lambda} \Rightarrow \phi = \frac{\Delta x}{\lambda} \times 2\pi$$

$$= \frac{\lambda/6}{\lambda} \times 2\pi = \frac{\pi}{3}$$

Let x intensity of each source

$$I = x + x + 2\sqrt{x^2} \cos\frac{\pi}{3} = 3x$$

& $I_0 = 4x$ ( $\because I_0 = x + x + 2\sqrt{x^2}\cos 0°$)

So $\frac{I}{I_0} = \frac{3}{4}$

18) Binding Energy = Δm × 931.5 MeV
= 3.5 × 931.5
= 3260.25 MeV

19) $X^{240} \rightarrow Y^{110} + Z^{130}$

$Q = (B.E)_{product} - (B.E)_{reactant}$

= [(7.8 × 110) + (8 × 130)] – [240 × 7.5]

= (858 + 1040) – 1800

= 1898 – 1800 = 98 MeV

20)

$d = \frac{t}{\cos r}\sin(i - r)$

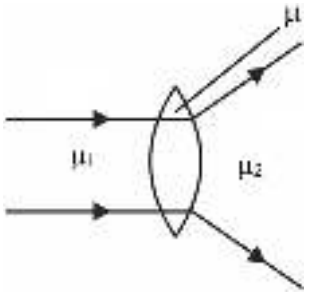
For small angle d = t (i – r)

Apply Snell's law sin i = μ sin r

i = μ r

$r = \frac{i}{\mu}$

$d = ti\left(\frac{\mu - 1}{\mu}\right)$




21)

μ = $\mu_1$; rays pass undiverted

As rays bend towards normal, $\mu_2 > \mu$

22) P– side of the diode should be at higher potential then n side.

23)

$\lambda = \frac{12400}{1.9}$ Å

= 6526Å

= 652.6 nm

24)

In conical pendulum, about vertical axis net torque is zero and hence angular momentum remains conserved.

25) Area of F-t graph = ΔP
$20 \times 2 - 1 \times 10 = m(v_f - v_i)$
$40 - 10 = 5(v_f - 0)$
$v_f = 6$ m/s

$$W = \frac{1}{2}m\left(v_f^2 - v_i^2\right)$$

$$W = \frac{1}{2} \times 5\,(36 - 0) \Rightarrow 90\,J$$

26) $$Y = \frac{F/A}{\Delta L/L}$$

$$\Delta L_{max} = \frac{\text{Elastic limit} \times L}{Y}$$

$$\Delta L_{max} = \frac{1.5 \times 10^8 \times 10}{12 \times 10^{10}} = 0.0125 \text{ m}$$

= 1.25 cm

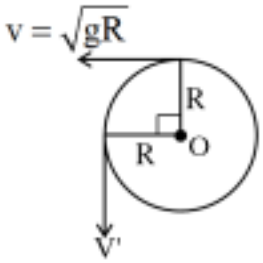
27)

$$F = \frac{GM.M}{d^2} = \frac{GM^2}{d^2}, F' = \frac{G\frac{M}{2}.\frac{3M}{2}}{\left(\frac{d}{2}\right)^2}$$

$$F' = \frac{3GM}{d^2} = 3F$$

Gravitational force does not depend on medium

28) Crossing with critical speed at top most point.



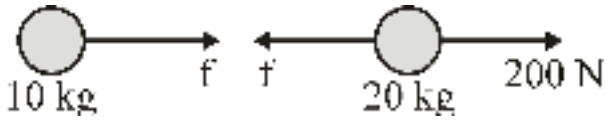

when string is horizontal,
$(v')^2 = v^2 + 2(g)(R) \Rightarrow (v')^2 = gR + 2gR = 3gR$

$$a_c = \frac{(v')^2}{R} = \frac{3gR}{R} = 3g$$

29)

It follows from the figure that the equations of motion are :

$200 - f = 20\ a_1$ and $f = 10\ a_2$




where $a_1$ and $a_2$ are the accelerations for 20 kg and 10 kg respectively.

But $a_2 = 12\ ms^2$

$\therefore$ $f = 10 \times 10 = 120$ N

$\therefore$ $a_1 = \frac{200-120}{20} = \frac{80}{20} = 4ms^{-2}$

30) Here, angular momentum is conserved, i.e., $L = I\omega$ = Constant. At A, the moment of inertia I is least, so angular speed and therefore the linear speed of planet at A is maximum.

31)

$L = mVr_{\perp}$

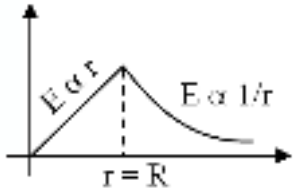
$= 5\ (3\sqrt{2})\left(4\frac{1}{\sqrt{2}}\right) = 60$ unit

32) $Mg\ (h + x) - F_{AV}X = 0 - 0$

$F_{AV} = Mg\left(\frac{h}{x} + 1\right)$

33)

Graph between induced electric field and distance.




34)

For stable equilibrium potential energy is minimum

$PE = - MB \cos\theta$

(1) $\theta = 90°$, $PE_1 = 0$

(2) $\theta = 90°$, $PE_2 = 0$

(3) $\theta = 180°$, $PE_3 = MB$

(4) $\theta = 0°$ $PE_4 = -MB$ (minimum)

So stable equilibrium

35) Voltage across inductor $V = -L\frac{di}{dt}$

$\because$ $i = i_0 \cos\omega t$

$$V = -L\frac{d(i_0 \cos \omega t)}{dt} = +Li_0\omega \sin \omega t$$

36)

α-particles: Heavy, doubly charged → produce dense ionization → maximum ionizing power
β-particles: Lighter, singly charged → moderate ionization
γ-rays: Neutral photons → least interaction → minimum ionizing power
Thus, the order of ionizing capacity is: α > β > γ
**Final Answer:** option (2)

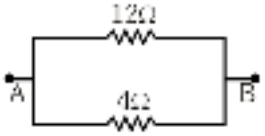
37) $$\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{140}{240 \times 0.7} \times 100$$
= 83.3%

38) at resonance power is max so $X_L = X_C$

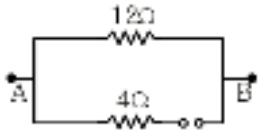
39)

$B_z = F(ky + vt)$
y + ive
So if is moving in –ive y direction

40) (i) $V_A > V_B$ ⇒ diode is in FB

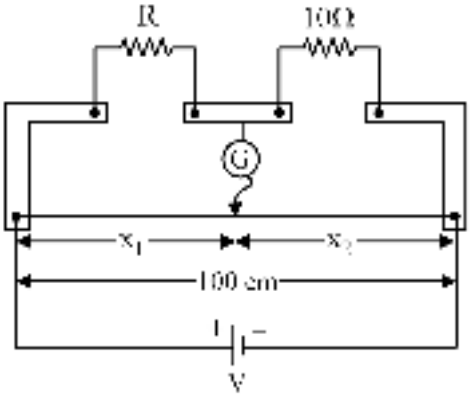



⇒
⇒ RAB = 3Ω
(ii) $V_A < V_B$ ⇒ diode is in RB




⇒
⇒ $R_{AB}$ = 12Ω

41) Statement-1 is true.
Work done in moving a test charge between any two point is, $W = q_0 (V_A – V_B) = 0$ (if $V_A = V_B$)
Statement-2 is false.
As at every point of equipotential surface, electric lines are ⊥ to this surface but are not mutually ⊥ to each other.




42)

Given, $\frac{x_1}{x_2} = \frac{4}{5}$

At balanced condition in meter bridge,

$\frac{R}{10} = \frac{4}{5}$

$\therefore$ R = 8Ω

Now, length of given wire whose resistance (8Ω) is 2 m.

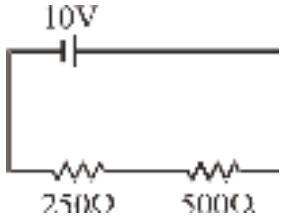
$\therefore$ 2m → 8Ω

$\therefore 1\Omega \rightarrow \frac{2}{8}m = \frac{1}{4}m = 2.5 \times 10^{-1}$ m

43)

$I_{min} = \frac{V}{R_{max}} = \frac{6}{10+20} = 0.2$ A

$I_{max} = \frac{v}{r_{min}} = \frac{6}{10+10} = 0.6$ A




44)

$I = \frac{10}{750}A$

$V = IR = \frac{10}{750} \times 250 = \frac{10}{3}V$

45)

Let m capacitor be in series & there are n such branches in parallel then

$C_{eq} = \frac{nC}{m}$ $V_{eq} = mV$

n = 2m m = 4

n = 8

Total number of capacitors = nm = 32

46) **Question Explanation:** Identify the electron configuration with the highest ionization energy.
**Concept:** Ionization Energy
**Solution:** Ionization energy decreases down a group and increases across a period. The $2p^5$ configuration (Fluorine) is in the second period and has a very high nuclear charge, giving it the maximum IE.
**Final Answer:** Option (4)

48) **Explanation :**
The question asks which cobalt complex ion will absorb light of the highest wavelength, which correlates to the smallest crystal field splitting caused by the ligands.

**Concept :**
Crystal Field Theory (CFT)

**Solution :**
The wavelength of absorbed light is inversely proportional to the crystal field splitting ($\Delta_0$). Weak field ligands cause smaller $\Delta_0$, leading to absorption of light at higher wavelengths (lower energy). Following the spectrochemical series ($F^- < H_2O < NH_3 < CN^-$), $F^-$ is the weakest field ligand among the options. Therefore, $[CoF_6]^{3-}$ will absorb light of the highest wavelength.

**Final Answer :**
Option (3)

49) **Generated by Allie**

**Problem Statement:** The question asks which ion among the given sulfur-oxygen polyatomic ions does not have a direct sulfur-sulfur (S-S) bond. The options are Sâ‚‚Oâ‚„Â²â», Sâ‚‚Oâ‚†Â²â», Sâ‚‚Oâ‚ƒÂ²â», and Sâ‚‚Oâ‚ˆÂ²â», and you need to identify the one without an S-S bond.

**Underlying Concept:** This problem is based on the structural chemistry of sulfur oxyanions. Some sulfur oxoanions contain direct sulfur-sulfur bonds, while others have sulfur atoms connected only via bridging oxygen atoms. Understanding the bonding and oxidation states in these ions helps determine the presence or absence of S-S bonds.

**Tips and Tricks:** Remember that thiosulfate (Sâ‚‚Oâ‚ƒÂ²â») and dithionite (Sâ‚‚Oâ‚„Â²â») commonly have S-S bonds, while polysulfate ions with more oxygen atoms tend to have sulfur atoms connected via oxygen bridges instead of direct S-S bonds.

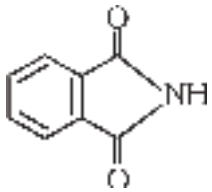
**Common Mistakes:** Assuming that all ions with two sulfur atoms must have an S-S bond, or not considering the role of bridging oxygens in the structure.

**Why Other Options Are Incorrect?:** Sâ‚‚Oâ‚„Â²â», Sâ‚‚Oâ‚ƒÂ²â», and Sâ‚‚Oâ‚ˆÂ²â» all feature direct S-S bonds in their structures, so selecting any of these as not having an S-S

bond is incorrect because they contradict known bonding patterns.

50)

$H_2S$ (basic)




51)
is the major product

52)

The transfer of hydride ion to the carbonyl group

53)

$AB_2 \rightleftharpoons AB + B$

t = 0 400 0 0

$t_{eq}$ 400–x x x

at eq$^m$ $P_T = 600$

$400 - x + x + x = 600$

$x = 200$

$$K_P = \frac{(x)(x)}{(400-x)}$$

54) c only

55)

b, c

56) 

$N_2 \xrightarrow{\text{electron}} N_2^-$

$(\sigma 2P_z)^2 < (\pi 2P_x)^2 \rightarrow (\sigma 2P_z)^2 < (\pi 2P_x)^1$

Diamagnetic Paramagnetic

$O_2 \xrightarrow{\text{electron}} O_2^-$

$(\sigma 2P_z)^2 < (\pi 2P_x^*)^1 \rightarrow (\pi 2P_x^*)^2 = (\pi 2P_y^*)^1$

57) Diamagnetic Paramagnetic

58)

A is true but R is false.

59)

6

60)

All of above

61)

$CH_3COOH$

62) T = 450 K  $\Delta H_r = 50\ kJ\ mol^{-1}$
$\Delta G_r = \Delta H_r - T\Delta S_r$.
For spontaneous reaction $\Delta G_r < 0$

63)

Radial nodes = n – □ – 1 = 3 – 2 – 1 = 0
Total nodes = n – 1 = 3 – 1 = 2

64)

$\Delta T_f = iK + M$ {i = 1}

$$0.4 = \frac{1 \times 5.12 \times 1 \times 1000}{MM \quad 50}$$

MM = 256

65) NCERT-XII, Pg. # 357

66) NCERT-XII, Pg. # 305

68)

All of these

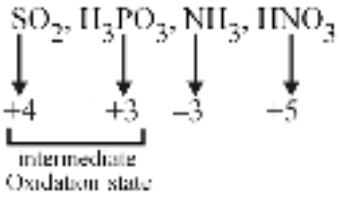
69) NCERT-2021, Part-1, Pg. # 113

70) NCERT-2021, Part-1, Pg. # 41

$K_H$ = Slope of the curve $= \frac{y_2 - y_1}{x_2 - x_1}$

$$= \frac{1000-0}{0.02-0} = 5 \times 10^4 \text{ torr}$$

71) NCERT-2021, Part-1, Pg. # 84




72)

73) (1) number of atom $= \left(\frac{5.6}{22.4}\right) \times N_A \times 3 = \frac{3}{4}NA$

(2) number of atom $= \left(\frac{0.5}{2}\right) \times N_A \times 2 = 0.5N_A$

(3) 18 ml of water = 18 g of water

number of atom $= \left(\frac{18}{18}\right) \times N_A \times 3 = 3N_A$

(4) number of atom $= \left(\frac{16}{48}\right) \times N_A \times 3 = N_A$

74)

$SiCl_4 + 4H_2O \xrightarrow{\Delta}$

75) a-P ; b-Q,R ; c-Q,R ; d-P,R,S

76)

$[Co(NH_3)_3Cl_3]$

77) The order of E.N. for boron family is.
B > Tl > In > Ga > Al

78)

a-S, b-R, c-Q, d-P

79) $C_4H_{10}(g) + 6.5O_2(g) \rightarrow 4CO_2(g) + 5H_2O(\square)$

$\Delta H = \sum (\Delta_f H)_{Product} - \sum (\Delta_f H)_{Reactant}$

$-2878 = [4\times(-393.5) + 5\ (-285.8)] - \left[(\Delta_f H)_{C_4H_{10}} + 6.5 \times 0\right]$

80)
**Question Explanation**: Calculating the half-life.
**Concept**: First-order reaction
**Solution**: The half-life of a first-order reaction is given by the formula t1/2=0.693/k. Given the rate and initial concentration, we can calculate the rate constant. The rate is k[A], and the initial concentration is 1M. So, k=0.6932×10−2. The half-life is 0.693/0.6932×10−2=100 minutes. **Calculation**: N/A **Final Answer**: 2

81) Trimerisation

$$i = \frac{1}{3}\alpha + (1-\alpha) = 1 - \frac{2\alpha}{3}$$

$$\frac{2\alpha}{3} = 1 - i$$

$$\Rightarrow \alpha = \frac{3}{2}(1-i)$$

82)

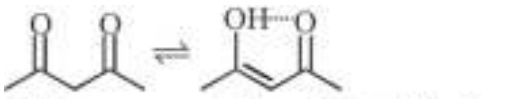
3.2

83) $XeF_6$ hydrolyses to form $XeO_3$ in a spontaneous reaction, so reverse is not feasible.

84) In $[Mn(CO)_6]$ ; EAN = 37.

85) Lactose is compound of β-D-galactose and β-D-glucose. The linkage is between $C_1$ of galactose and $C_4$ of glucose is called β-glycosidic linkage.

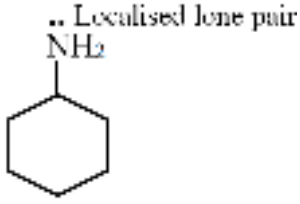
86)




2,4 Pantane dione

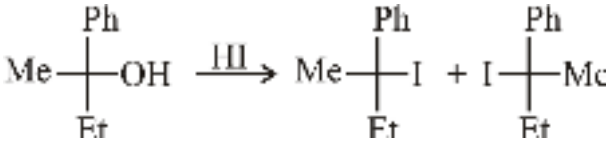
(enol form 76%, stabilised by intramolecular hydrogen bonding & extended resonance

87)

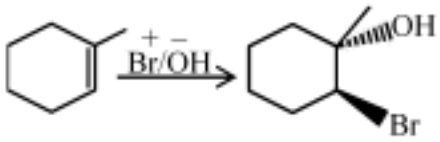



88)

$S_N1$ leads to racemisation in the following question :-



89) **Question Explanation:** Identify the correct major product of the given reaction.
**Concept:** EAR : $X_2 + H_2O$
**Solution:**



Anti addition through non classical carbocation

**Final answer:** option (2)

90) ZnS is white ; CdS is yellow; CuS & HgS are black. However only $Hg^{+2}$ is reduced by $SnCl_2$ but not $Cu^{+2}$.

BIOLOGY

92)

Sexes are seperate

93) NCERT, Pg. # 44, 4.2.7

94)

Three

95)

Lining of body cavity

96) NEW NCERT XI Page no. 81 Para 7.2.2

97)

NCERT Page No. # 188

98)

Pleural membrane

101) NCERT (E) Pg. # 331

102) NCERT Page # 334, 335, 336

103)

**Explain Question:** The question asks which of the listed hormones is **not a gastrointestinal hormone**.

**Concept:** Gastrointestinal hormones

**Solution:** The correct answer is Option 4.
**Erythropoietin** is produced by the **kidney** and stimulates **red blood cell production**, not a gastrointestinal hormone.

A. Gastrin, Secretin, and Cholecystokinin are all **GI hormones** regulating digestion.

**Final Answer:** Option 4

104)

A = Q, B = T, C = S, D = R

105)

Bowman's capsule + PCT + DCT + Loop of Henle

106)

**Question Explanation:** The question asks to identify muscles that show striations but function involuntarily.

**Concept:** Cardiac muscle

**Solution:** Cardiac muscle (heart muscle) is striated in structure but involuntary in function.

**Final answer: Option 2**

107)

Cervical → Thoracic → Lumbar → Sacral → Coccyx

108) NCERT Pg # 43
Assertion (A) describes only list of mole accessory ducts and reason (R) describes only structural arrangement of epididymis and vasa-deferens but it is not explain why they are called male accessory ducts.

कथन (A) में केवल नर जनन सहायक नलिकाओं की सूचि दि गई है। तथा कारण (R) केवल एपिडीडाइमिस एवं वसा डेफरेंस के संरचनात्मक व्यवस्था को वर्णित करता है लेकिन इनको नर जनन सहायक नलिकायें कहा जाये इसकी व्याख्या नहीं करता है।

109)

Falls, AV valves

110) NCERT Pg. # 288

111)

NCERT XII Pg # 49, Fig. 3.8

112) NCERT, Pg # 48,49
Oogenesis already started during foetal life

113) NCERT ($XII^{th}$) Pg. # 36

114)

NCERT - XII Pg. No 45

115)

**Question Explanation:** Match the column.

**Concept :** Menstrual Cycle

**Solution :**

A. Menstrual phase - Breakdown of endometrium
B. Follicular phase - Fully mature graafian follicle
C. Ovulatory phase - Release of ovum
D. Luteal phase - Corpus luteum

**Final Answer :** option (3). A-ii, B-iii, C-iv, D-i

116) XII NCERT Pg # 77

117) **Explain Question:** The question asks to analyse the given statement I and statement II and select the correct answer.

**Concept:** Dihybrid Cross.

**Solution:**

Statement-I : Based upon observations on dihybrid crosses, Mendel proposed the law of independent assortment.

Statement-II : In F2 generation of dihybrid cross Mendel obtained 75% plants having round seed.

Both statements I and II are correct.

So, the correct answer is: 4.

**Final answer:** 4.

118)

NCERT-XII, Pg. # 129

119) NCERT-XII, Pg. # 141-142

120)

NCERT XII_Pg. No. 118

121) NCERT Pg. # 115

122)

**The correct answer is Option 3: Preventing ovulation.**
Contraceptive oral pills work by altering the hormonal balance in the body, specifically by inhibiting the release of luteinizing hormone (LH) and follicle-stimulating hormone (FSH). This prevents the ovary from releasing an egg (ovulation), thereby reducing the chances of fertilization and pregnancy. Additionally, these pills may thicken cervical mucus, making it harder for sperm to reach the egg.

123) NCERT Page. No. 201

124) NCERT Pg. # 230

125) NCERT Pg. # 242

126) NCERT Pg. # 212, 214

127) NCERT Pg. # 207, 227, 228

128) **Explain Question:**Chromatid number per chromosome at metaphase?

**Concept:** Metaphase

**Solution:** In metaphase of mitosis → each chromosome = 2 chromatids.

In metaphase-I of meiosis → homologous chromosomes pair, but each chromosome still has 2 chromatids.

At metaphase, whether in mitosis or meiosis-I, every chromosome has two sister chromatids.

**Final answer: Option 1**

129)

NCERT Pg. # 96

130)

A-II, B-IV, C-I, D-III

131) NCERT-XI, Pg # 123

132)

NCERT XII Pg. # 205

133) NCERT Pg. # 191

134) NCERT Pg. # 219

135) NCERT Pg. # 222, 223

136)

NCERT-XII, Pg. # 198

137)

Presence of more than one restriction sites of a restriction enzyme

138)

A → The passage through which movement of RNA and protein molecules take place in both direction

139) NCERT Page No. 8 (Table 1.1)

140) NCERT-XI Pg. # 14, 15

141)

NCERT XI - Pg. # 33

142) NCERT XI Pg. # 140, 146, 149

143)

NCERT-XI Pg. # 146

144) NCERT Pg. # 140

145) NCERT-XI Pg. # 160

146)

NCERT XI, Pg. # 157

147) NCERT Pg. # 174,176,177,178

148) NCERT-XI, Pg. # 66

149) NCERT Pg. # 69(E), 68(H)

150) NCERT, Pg. # 75

151) NCERT, Pg. # 76

152) NCERT Pg. # 7

153) NCERT XII Pg. No. # 18, 19

154) NCERT-XI, Pg. # 176

155) NCERT Pg. No. # 66

156) NCERT-XI, Pg. # 26

157) NCERT, Pg. # 21

158) NCERT Pg. # 109

159) NCERT XI Pg.- 6, 7

160) XII NCERT Page No. # 80, 81

161) NCERT, Pg. # 107

162) NCERT Pg. # 113

163) NCERT Pg. # 99

164) NCERT, Pg. # 64

165) Module No. 9 Page. No.110

166)

lane - 1

167) NCERT Pg. # 65, 67, 69

168) NCERT Pg. # 65, 67

169) NCERT Pg. # 61

170) NCERT Pg. # 88

171) NCERT, Pg. # 81, 82

172) NCERT Pg. No. 96

173) NCERT Pg. # 62

174) NCERT, Pg # 69

176)

Plants

177)

Translation

178) NCERT Pg # 210, 214

179) NCERT Pg # 152

180)

A-peptide and B-peptide peptides are present in mature insulin.